

SESSION 6 — DIMENSIONAL MODEL DESIGN

Milestone B3 · Grain, facts and dimensions, schema on paper

BDAT04 Fundamentals of Data Warehousing · Saturday 26 September 2026 · Three hours

How this activity works

- Eight items, in three parts. Work them in order — each one uses what the one before it produced.
- Every item has the same two boxes. ANALYSIS is what you found or built. REFLECTION is what it means. Both are marked, and the reflection is worth more than you expect.
- You may use Power BI, your notes, your earlier queries and your feedback letters. Part C is done on paper.
- This activity IS milestone B3. There is no separate submission.

A reflection is not a summary.

Do not tell me what you did — the analysis box already says that. Tell me what it cost, what it would have broken, or what you now see that you did not see before. Two or three honest sentences beat a paragraph of restatement.

Dataset: Kape Tayo, as it stands in your own Power BI workspace. Nothing new is issued today.

PART A — GRAIN

Grain is the single sentence that says what one row of a fact table means. It is the first thing a designer declares and the most expensive thing to get wrong.

ITEM 1 • What is one row of tblSales?

about 20 minutes

What to do

1. Load tblSales. Count the total rows.
2. Count the DISTINCT values of SaleID.
3. Compare the two numbers. If they are equal, each SaleID appears once. If they differ, some appear more than once.
4. Look at three actual rows. What does each one describe — a whole sale? a single product within a sale? a day of sales?

ANALYSIS

State the grain in ONE sentence, beginning exactly: “One row of tblSales is ...” Then give the two counts you used.

One row of tblSales is a product's net sale on a specific Branch on a particular day.

Counts used

- Net sales
- Branch
- Date
- Product

REFLECTION

If your grain sentence is wrong, every number built on this table is wrong and nothing warns you. Which of the two counts would have exposed the error — and what would you have concluded if you had only looked at the rows?

If the BranchID data is incorrect, the other counts would not have a specific identity for what they were categorized as.

- I would have concluded that a certain product was sold to a certain branch on a particular day.

ITEM 2 • What is one row of tblInventory?

about 20 minutes

What to do

5. Load tblInventory. Count the rows.
6. List the distinct values of MovementType and count the rows in each.
7. Decide what a single row records. Is it a stock level, or a stock movement? Those are not the same thing.
8. Check whether any column could serve as a unique identifier for the row.

ANALYSIS

State the grain in ONE sentence, beginning “One row of tblInventory is ...” List the MovementType values with their counts.

One row of tblInventory is a dispatch of a certain number of products to a specific Branch on a particular day.

MovementType Values

- Adjustment – 23 counts
- Delivery – 1476 counts
- Transfer Out – 19 counts
- Wastage – 23 counts

REFLECTION

tblSales and tblInventory are both fact tables, and their grains are different. Write one sentence explaining why you cannot put them in the same table, even though both are about products at branches over time.

Both tables have a different story behind the data they project. tblSales talks about a sale of a certain product at a specific branch, whereas tblInventory talks about the number of products stocked and delivered at a specific branch.

ITEM 3 • The grain test

about 25 minutes

What to do

9. Take this business question: “How much revenue did each branch earn in March 2026?” Using only your declared grain for tblSales, decide whether the table can answer it. Show the grouping you would use.
10. Now take this one: “What was the average number of items per customer visit?” Decide whether your declared grain can answer it.
11. Do not build the answer. Decide whether the grain permits it, and show why.

ANALYSIS

For each question: ANSWERABLE or NOT ANSWERABLE, and the one-line reason that follows from your grain statement.

“How much revenue did each branch earn in March 2026?” – Answerable

- Because my grain statement talks about a product’s sales performance at a branch on a specific date.

“What was the average number of items per customer visit?” – NOT ANSWERABLE

- This statement could not be answered since it talks about a customer’s spending habits, whereas my statement focuses on a product’s performance on a specific Branch.

REFLECTION

One of these two questions is blocked by the grain, not by missing data. Explain the difference between “we do not have that data” and “the grain of this table cannot express that” — and say which one a business user would find harder to accept.

The difference between the two statements is the availability of data presented by each grain; tblSales talks about a product’s performance, whereas tblInventory talks about a product's movement. This means that each set of data tells a specific story that only a particular set of questions can answer, but it does not necessarily mean the absence of the said data.

PART B — FACTS AND DIMENSIONS

A fact table holds the things you measure. A dimension table holds the things you slice by. Most modelling errors are really classification errors made early and never revisited.

ITEM 4 • Classify every column in tblSales

about 25 minutes

What to do

- 12. List every column in tblSales.
- 13. Put each one into exactly one of three buckets: MEASURE (something you would sum or average), KEY (something that points at another table), DESCRIPTIVE ATTRIBUTE (something you would filter or group by, but never sum).
- 14. Two columns will be arguable. Decide anyway, and mark them with an asterisk.

ANALYSIS

A three-column table: Column | Bucket | Why. Every column in the table must appear exactly once.

COLUMN	BUCKET	WHY
SaleID	DESCRIPTIVE ATTRIBUTE	Each of these data points is an identifier for a certain sale.
Date	DESCRIPTIVE ATTRIBUTE	Each of these data points is an identifier for a certain sale.
BranchID	KEY	This represents a specific branch
ProductID	KEY	This represents a specific product
CustomerID	KEY	This represents a particular customer
Quantity	MEASURE	This represents the number of products sold
UnitPrice	MEASURE	This represents the product's price
Amount	MEASURE	This represents the net sales
PaymentMethod	DESCRIPTIVE ATTRIBUTE	This represents the mode of payment

REFLECTION

Find a column you could technically sum but never should. Name it and say what the sum would mean — and why producing that number would mislead a reader.

A column that I would technically sum is the column "Quantity." This would represent the total number of products sold. However, this would entirely mislead a reader since the products sold, and its corresponding branch is not available with this data alone.

ITEM 5 · Classify the seven tables

about 25 minutes

What to do

- 15. List the seven Kape Tayo tables.
- 16. Classify each as FACT, DIMENSION, or NEITHER.
- 17. For each FACT, write its grain in one line (you already have two of them).
- 18. For each DIMENSION, name the column that joins it to a fact table.

ANALYSIS

A four-column table: Table | Fact / Dimension / Neither | Grain (facts only) | Join key (dimensions only).

TABLE	FACT / DIMENSION / NEITHER	GRAIN	JOIN KEY
tblSales	FACT	<u>One row of tblSales is a product's net sale on a specific Branch on a particular day.</u>	
tblBranches	DIMENSION		BranchID
TblInventory	FACT	<u>One row of tblInventory is a dispatch of a certain number of products to a specific Branch on a particular day.</u> <u>MovementType Values</u>	
tblCalendar	NEITHER		
tblProducts	DIMENSION		ProductID
tblEmployees	NEITHER		
tblCustomers	DIMENSION		CustomerID

REFLECTION

Name the table you found hardest to classify and say what made it hard. A hard case is not a failure — it usually means the business itself has not decided something.

The table that I found hardest to classify is tblCustomers, since its data independently describes the customer's spending behavior; however, it does not record the walk-in customers' behavior. But it also relates to branches and their sale performance.

ITEM 6 • The hard case — tblEmployees

about 20 minutes

What to do

19. Open tblEmployees. Read its columns.
20. Ask: which fact table has a column that points at an employee? Check tblSales and tblInventory.
21. Decide: is tblEmployees a dimension in THIS model, or is it a table that belongs to a different model entirely?
22. If it is a dimension, name the fact table it would serve and the key it would use. If it is not, say what it would take to make it one.

ANALYSIS

Your decision, in writing, with the evidence you checked.

tblEmployees is a table that belongs to a different model entirely. All the other tables tell a common story, which is the sales performance of the branches for the available products that KapeTayo sells. Whereas the table tblEmployees only talks about the employees' employment status, wage, role, hire date, designated branch, and nothing else that could help enhance the model's credibility.

REFLECTION

A dimension nobody joins to is not a dimension — it is a table sitting in your file. What would you say to a manager who insists it belongs in the model because “we have the data”?

I would say that this data is important if you want to build a model analyzing the employees' performance; however, for this specific model where we are analyzing the business's sales performance, it is not necessary to include it in the model.

PART C — SCHEMA ON PAPER

You draw the schema before you build it, because paper forces you to decide what the software would quietly decide for you.

ITEM 7 • Draw the schema on paper

about 30 minutes

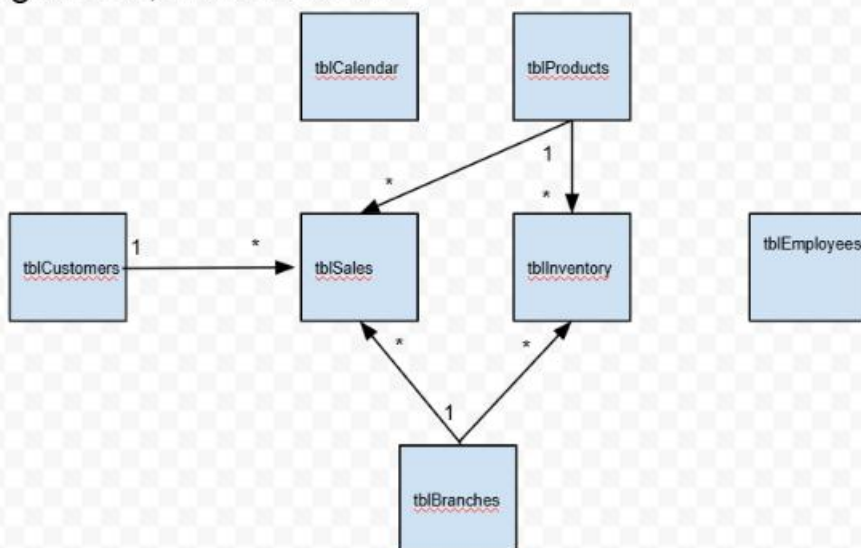
What to do

23. On PAPER — not in Power BI — draw your model.
24. Put your fact table or tables in the centre. Put the dimensions around them.
25. Draw a line for every relationship. Write the key name on each line.
26. Mark the SIDE of each relationship: which table is ONE, which is MANY. Use a 1 and a *.
27. Label your diagram with your name and photograph it.

ANALYSIS

A photograph of your drawing, inserted below or submitted alongside this sheet.

Lagdamen, Lawrence M.



REFLECTION

You drew this before building it. Why does the syllabus ask for paper first? Answer from what happened while you were drawing — something you had to decide on paper that the software would have decided for you.

In order for me to understand how each of these tables are correlated with each other and their corresponding use/identity for the model.

ITEM 8 • The dimension that serves both

about 15 minutes

What to do

28. Look at your drawing. Find the dimension or dimensions that connect to BOTH fact tables.
29. Name them.
30. Now imagine you had built two separate copies — one for Sales, one for Inventory.

ANALYSIS

The name of the shared dimension, and the two fact tables it serves.

- tblProducts
- tblBranches

REFLECTION

Describe one specific report that would go wrong if Sales and Inventory each had their own private copy of that dimension. Be concrete: name the report, and say what number would be wrong.

It would be the number of products in each branch. There would be a different number of products on the branches if they have access to a different set of data for the given tables.

B3 — FORMAL SUBMISSION BLOCK

Copy your final answers here. This is the page marked against the syllabus milestone. Everything in it should already exist somewhere in Items 1 to 8 — if it does not, go back.

1 • Grain statement, one per fact table

Written as a full sentence. "One row of ... is ..."

One row of tblSales is a product's net sale on a specific Branch on a particular day.

One row of tblInventory is a dispatch of a certain number of products to a specific Branch on a particular day.

2 • Fact tables, with their measures

tblSales — Measures a product's performance on a specific branch on a particular day.

tblInventory — Measures a the dispatch movement of a certain number of products to a specific Branch on a particular day.

3 • Dimension tables, with the key each one joins on

tblBranches — BranchID

tblProducts — ProductID

tblCustomers — CustomerID

4 • Relationships — one line each, in the form DIMENSION 1 — * FACT (key)

tblProducts 1 ---- * tblInventory

tblProducts 1 ---- * tblSales

tblBranches 1 ---- * tblInventory

tblBranches 1 ---- * tblSales

tblCustomers 1 ---- * tblSales

5 • One design decision you are not certain about, and what you would need to resolve it

I am not completely certain whether tblEmployees should be included in this dimensional model. To resolve it, I would need to know whether the business wants to analyze the employees individual sales performace.

Schema drawing attached / photographed: YES Name: Lagdamen, Lawrence M.

How this is marked

Criterion	Weight
Grain statements are correct, complete and written as sentences	25
Facts, dimensions and keys classified correctly, with the hard cases defended	25
Schema drawing — complete, keyed, sides marked	20
Reflections — honest, specific, and about consequence rather than restatement	25
Submitted complete and on time	5

An item with a correct analysis and an empty or restating reflection earns roughly half of what it could. That is deliberate. The analysis shows you can operate the tool; the reflection shows you understand what you just did.

— Sir JB